

LONG ISLAND UNIVERSITY · LEWYT COLLEGE OF VETERINARY MEDICINE

DVM 612 · PRINCIPLES OF SURGERY

# Preoperative Evaluation, Patient Preparation & Postoperative Care

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Lecture | 37 slides | 60 minutes

Exam, labs, then you write the ASA. Then prep. Then 24 hours of monitoring and complication management.

Required: Fossum 2018; Hendrickson & Baird 2013.

Also used: ECFVG CPE Manual of Administration 2026, Anesthesia and Surgery.

# Learning objectives

- ▶ Assign American Society of Anesthesiologists (ASA) status after today's PE and labs. Correct deficits. Write premed, induction, MAC, tube, bag, and fluid rate.
- ▶ If albumin is low: find the cause, delay elective if you can, and do not flood with crystalloid.
- ▶ Prep for aseptic surgery: clip, dirty scrub, OR attire, full scrub, drape, instrument table, closed glove.
- ▶ Name a break in asepsis and correct it, before or after the incision. Classify the wound: clean, clean-contaminated, contaminated, or dirty.
- ▶ Write the postop plan, the surgical report, and 24-hour emergency criteria.

Exam → labs → ASA → ready for prep → clip, dirty scrub, OR, full scrub, drape, instruments → first 24 hours.

# Prepare the patient and yourself

## Causes of SSI

Hair, skin flora, hypothermia, poor hemostasis, dead space, and breaks in asepsis.

## Client-owned patients

Elective OHE patients go home to the owner after recovery. Documentation and the discharge conversation are part of surgical care.

## Set before you cut

Analgesia, antibiotics, temperature management, and client expectations are set in the preoperative period.

## Surgeon responsibilities

- ▶ Prepare the patient for a surgical procedure.
- ▶ Prepare yourself for a surgical procedure.
- ▶ Perform the procedure. Then monitor the first 24 hours: a weak ligature or a weak linea is a technical failure you must recognize (CPE MOA 2026 Surgery).

# Halsted's principles

## 1. Gentle tissue handling

Prep trauma, clipper burn, and crushing towel clamps are tissue handling.

## 2. Meticulous hemostasis

Preop coagulopathy; postop hemorrhage is a 24-hour emergency.

## 3. Preserve blood supply

Place clamps and bandages so skin perfusion is preserved.

## 4. Strict asepsis

Patient preparation, surgeon preparation, and draping.

## 5. No tension

Plan incision location and closure now, not after the fascia splits.

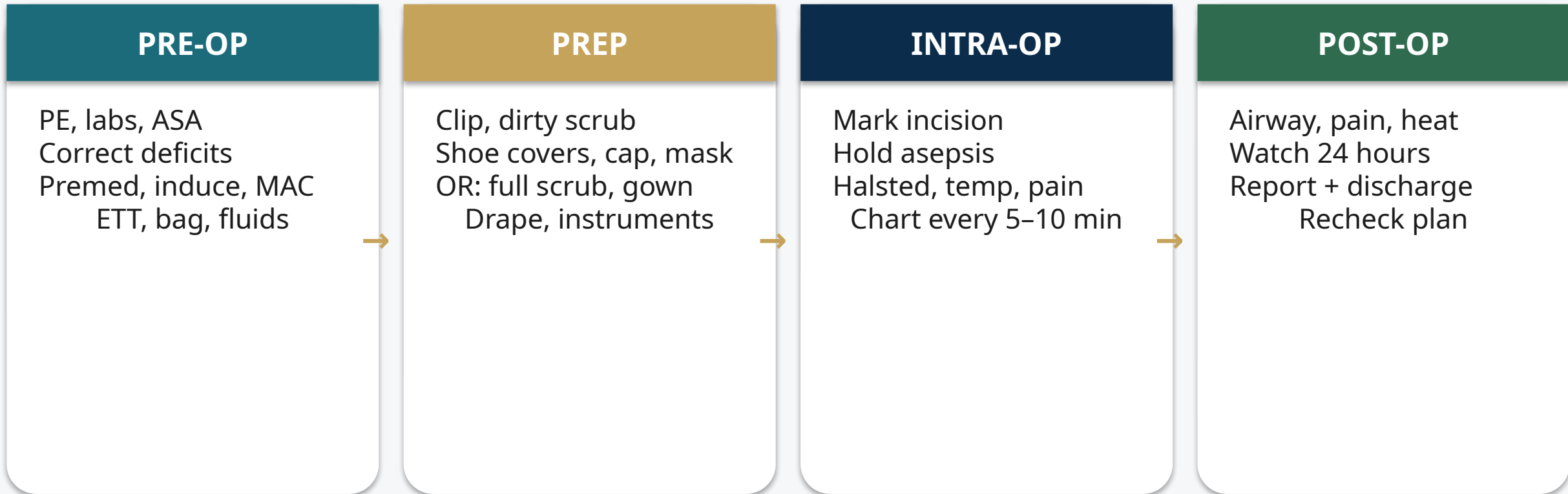
## 6. Accurate apposition

Postop incision care protects the apposition you create.

## 7. Obliterate dead space

Plan dead space now: sutures, bandage, or a drain with a written pull plan.

# The perioperative continuum



## Anesthesia clock · ready for surgical prep

From the preoperative exam: airway in and secured, machine on, patent IV running, surgical plane, monitoring started. Then call for prep (CPE MOA 2026 Anesthesia).

## Surgery clock · last skin suture

From the start of patient preparation to the last skin suture, including re-gowning. Then 24 hours of monitoring and complication management (CPE MOA 2026 Surgery).

# Preoperative physical examination

## T • HR • RR

Write the numbers. Conscious dog: HR 70–140, RR 20–40. Cat: HR 145–200, RR 20–40 (Fossum).

## mm / CRT

Mucous membrane color and capillary refill. Pink vs pale vs injected. CRT in seconds.

## Heart and lungs

Auscultate both sides. Record any murmur, arrhythmia, or crackles before you choose drugs.

## Hydration + site

Skin tent, tacky mm. Look at the surgical site: pyoderma, heat, pain. Abdomen if you might open it.

## Minimum labs on the anesthesia record • PCV, TP, BUN (CPE MOA 2026 Appendix 3)

### PCV

Oxygen-carrying capacity. Anemia vs hemoconcentration. With TP: dehydration vs blood loss.

### TP

Protein / oncotic pressure. High with PCV: dehydration. Low: loss, liver, or dilution. Changes fluids and healing.

### BUN

Kidney, hydration, GI bleed. Azotemia: skip NSAIDs, rethink fluids, maybe do not cut today.

# American Society of Anesthesiologists (ASA)

| ASA status | Definition                                              | Example                                       |
|------------|---------------------------------------------------------|-----------------------------------------------|
| Status 1   | Normal healthy patient                                  | Healthy patient for neutering                 |
| Status 2   | Patient with mild systemic disease                      | Well compensated mild mitral degeneration     |
| Status 3   | Patient with moderate systemic disease                  | Cat with CKD, IRIS stage 3                    |
| Status 4   | Severe systemic disease; constant threat to life        | Hemoabdomen, bleeding splenic mass            |
| Status 5   | Moribund; not expected to survive without the operation | Hypotensive, hypothermic, obtunded septic cat |

E = emergency. Status 3 is moderate and compensated. Status 4 is uncompensated and a threat to life. Today's PE and labs write the number. A normal CBC does not write Status 1.

# MoMo: today's PE



SF DSH, 6 yr, 4.25 kg, BCS 5/9. Acute vomiting and lethargy this morning. Two vomits (food, then liquid). Construction at home. T 98.0 °F, HR 200, RR 30, mm pink tacky, CRT <2 s, QAR. Do not write the ASA yet.

## Why this looked surgical

Possible FB or toxin was on the list. Mildly enlarged abdomen. That is how cats get booked for an exploratory.

## Abdomen and the rest of PE

Mildly enlarged abdomen on palpation. Rectal not examined. The PE did not prove a foreign body.

## Also on PE

Eyes, ears, nose normal. Oral pink, CRT <2. Lungs clear, eupneic. No murmur, regular rhythm. Ambulatory ×4. LN and skin WNL. Neuro: normal gait, CNs intact. MCS 3/3. Generalized small stature.

# MoMo: labs. Write the ASA

## HCT

44% (28–50)

## TP (total protein)

5.1 g/dL

## BUN / creatinine

BUN 48.7 → 100 (15–32). Cr 3.0 → 4.71.

## Out of range

WBC 26 (3–16).  
Neutrophils 24 (4–11). Na 142. BE –7.4.  
Glucose 136. Phos high; no number printed.

**Quiz. Assign ASA. Add E if emergency. Write it before you hear the key. AUS: right kidney fluid-filled, non-functional; left kidney reduced corticomedullary architecture. HCT in range does not write Status 1.**

## 1. Status?

Number, and E or not.

## 2. Why not Status 3?

Is this still compensated CKD, or a constant threat to life?

## 3. Cut today?

Exploratory on vomiting and an enlarged abdomen?

# Willie: today's PE

MN Cavalier, 6 y 11 mo, 13.7 kg. Acute ataxia and disorientation ~1 hour. Found on the floor; fell off the couch twice. Cytopoint. T 100.8 °F, HR 132, RR 52, mm pink, CRT 2 s, dull/quiet. Do not write the ASA yet.

## Left ear (AS)

Brown and bloody discharge.  
Pedal reflex at the pinna base.  
Canal patent. Cartilage hardened. Tympanic membrane visible but swollen.

## Vestibular

Head tilt / leaning toward the left (infected ear, AS). Stumbling. Circling or tight turns to the left. Nystagmus: rapid left, slow right.

## Wheelbarrow / CP

Wheelbarrow: left intact, right absent. Prolonged proprioception, right limbs. Knuckling on the right limbs when walking. Limbs slip.

Also on PE: Grade II/VI left systolic murmur. Lungs clear, eupneic. Heavy tartar. Yellow crusts on the nasal philtrum. Eyes: PLR present. Next: today's chemistry and CBC. Then you write the ASA. Owner identifiers stay off these slides. Do not use a drug list on this dog.

# Willie: labs. Write the ASA

## HCT

47.5% (36.9–60.0)

## TP

6.4 g/dL (5.5–7.6)

## BUN / creatinine

BUN 11.9 (9.0–29.0). Cr  
0.7 (0.4–1.4).

## Out of range

Phos 6.0 (1.9–5.0).  
Glucose 130 (75–  
125). CBC in range.

**Quiz. Assign ASA. Add E if emergency. Write it before you hear the key. HCT in range does not write Status 1.**

### 1. Status?

Number, and E or not.

### 2. Why not Status 1?

What on today's PE is not a normal healthy patient?

### 3. Why not Status 4?

Is this a constant threat to life today?

# Pre-anesthetic assessment

PCV / TP: EDTA tube → spin hematocrit → read PCV → snap the capillary tube → refractometer TP. BUN: chemistry or Azo stick. Request, record, interpret. Then ASA. Then premed IM/SQ. Not before the exam.

| Field       | MoMo (quiz)                                               | Healthy Lab OHE           | Willie (quiz)                                     |
|-------------|-----------------------------------------------------------|---------------------------|---------------------------------------------------|
| Weight      | 4.25 kg                                                   | 25 kg                     | 13.7 kg                                           |
| T · HR · RR | 98.0 °F · 200 · 30                                        | normal TPR                | 100.8 °F · 132 · 52                               |
| mm / CRT    | pink, tacky / <2 s                                        | normal                    | pink / 2 s                                        |
| Other PE    | Mild abdominal enlargement<br>Ambulatory ×4. No FB proven | Elective OHE<br>normal PE | AS otitis. Head tilt L<br>Circling L. R knuckling |
| PCV / HCT   | HCT 44%<br>(28–50)                                        | within reference          | HCT 47.5%<br>(36.9–60.0)                          |
| TP          | 5.1 g/dL<br>(total protein)                               | within reference          | 6.4 g/dL<br>(5.5–7.6)                             |
| BUN         | 48.7 → 100<br>Cr 3.0 → 4.71                               | within reference          | 11.9 · Cr 0.7<br>Phos 6.0 · Glu 130               |
| ASA         | write it                                                  | 1                         | write it                                          |

# Laboratory values inform risk

| Finding   | Typical healthy adult | Repeat / stabilize first                         |
|-----------|-----------------------|--------------------------------------------------|
| PCV       | Within reference      | Anemia: use severity and chronicity (Grubb 2020) |
| TP        | Within reference      | High with PCV: dehydration. Low: loss / dilution |
| BUN       | Within reference      | Azotemia: skip NSAID, rethink fluids and the cut |
| Potassium | Within reference      | >6.0 mEq/L (Grubb 2020: correct first)           |
| Albumin   | Within reference      | <2.0 g/dL: edema, poor healing (Pardo 2024)      |

MOA minimum is PCV, TP, BUN. A normal CBC does not assign Status 1. Azotemia can cancel the cut. Low albumin is a deficit, not an ASA number.

# Anesthetic drug protocol

## 1. Premed after ASA

IM or SQ. Not before the exam.  
Hospital: DKT or BAA. Same columns, hospital doses.

## 2. Induce IV to effect

Alfaxalone 10 mg/mL. Draw 5.5 mL.  
Give until you can intubate.

## 3. Iso or sevo to MAC

Premed, if you add it, lowers MAC and lowers alfaxalone.

### Anesthetic drug protocol

25 kg · ASA 1 · elective OHE

|           | Drug                 | Concentration | Dose         | Volume          | Route          |
|-----------|----------------------|---------------|--------------|-----------------|----------------|
| Analgesic | Carprofen (Rimadyl)  | 50 mg / mL    | 4.4 mg / kg  | 2.2 mL (110 mg) | SQ ~2 h before |
| Induction | Alfaxalone (Alfaxan) | 10 mg / mL    | IV to effect | draw 5.5 mL     | IV             |
| Inhalant  | Isoflurane           | to MAC        | to effect    | vaporizer       | inhalant       |

|           |                                                     |                                                                                                                                                                                                                                                                                                                  |
|-----------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IV fluids | <b>Name:</b> LRS<br><b>Rate:</b> 125 mL/hr (25 × 5) | <b>Why these drugs</b><br>Carprofen: 25 × 4.4 = 110 mg. 110 / 50 = 2.2 mL SQ (Rimadyl).<br>Alfaxalone: draw 25 × 2.2 = 55 mg = 5.5 mL. Give IV to effect (Alfaxan).<br>2.2 mg/kg is the unpremedicated field-study mean, not a fixed bolus.<br>Cat OHE: Onsior 2 mg/kg SQ. Do not stack an NSAID with a steroid. |
|-----------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Sedation quality 1-5        |
|-----------------------------|
| 1 2 3 4 5                   |
| Circle circuit              |
| Rebreathing Non-rebreathing |

| Induction quality 1-5    |
|--------------------------|
| 1 2 3 4 5                |
| ETT · bag (if intubated) |
| 10 mm 2 L circle         |

| Circle inhalant             |
|-----------------------------|
| Isoflurane Sevoflurane      |
| Fresh-gas flow              |
| 2-3 L/min, then ≥ 0.5 L/min |

# Correct deficits before you cut

## Correct these first

Grubb et al. 2020, before anesthesia:

Dehydration / hypovolemia

K<sup>+</sup> >6.0 mEq/L

pH <7.2

Glucose <60 mg/dL

Hypothermia <99 °F

Anemia (severity, chronicity)

Life-threatening arrhythmia

CHF, cyanosis, pneumothorax

Oliguria / anuria

Uncontrolled pain or anxiety

## If albumin is low

Find the cause: GI loss, kidney, liver, exudate, starvation.

Delay elective surgery if you can treat and feed.

<2.0 g/dL: edema, poor perfusion, poor wound healing (Pardo 2024).

Plasma: about 20–25 mL/kg to raise albumin 0.5 g/dL. Often impractical.

Species-specific albumin if severe and you must operate. Human albumin can cause anaphylaxis.

Do not flood with crystalloid when the problem is oncotic pressure.

## If you still operate

Delay elective for pyoderma over the site.

Emergency: resuscitate volume, then cut. You cannot wait for albumin to normalize in GDV or septic abdomen.

Gentle tissue handling. Obliterate dead space. Warm the patient. Nutrition as soon as the gut allows.

Rising creatinine that threatens life: cancel the elective or exploratory cut.

# Fasting and informed consent

## Fasting (Grubb et al. 2020 AAHA)

- ▶ Healthy adult dog/cat: food 4–6 h; water until premedication.
- ▶ Some hospitals still use 8–12 h NPO. Teach aspiration versus hypoglycemia.
- ▶ Neonates / <2 kg: food fast no longer than 1–2 h.
- ▶ Brachycephalics: same fasting as the adult or neonate row. Pre-oxygenate 3 min with 100% O<sub>2</sub> (Grubb 2020). Extubate when they can protect the airway, not at the first swallow.
- ▶ Ask what was eaten this morning. Clients feed 'just a biscuit.'
- ▶ Diabetics: write the insulin dose and a small meal with anesthesia before drop-off.

## Consent, then file it

- ▶ Procedure name in plain language, and the reason.
- ▶ Benefits, alternatives including medical management, and what happens if we wait.
- ▶ Material risks in one sentence each: anesthesia death, hemorrhage, infection, dehiscence. After OHE, name urinary incontinence.
- ▶ Estimate: what is included and what is not.
- ▶ Resuscitation code / DNR before induction.
- ▶ File the signed form. Then start induction.

# ASA, tube, bag, and fluid rate

## Endotracheal tube

$\text{ID mm} = (\text{kg} / 4) + 3.5$

Dog start. ±0.5 mm ready. Largest through arytenoids (Grubb 2020). Cat: 3.0–4.5 mm (not the dog formula).

## Reservoir bag

$\text{mL} = \text{kg} \times 60, \text{ round UP}$

Round up to 0.5, 1, 2, or 3 L. NRC if <3–5 kg. Circle if larger. Leak-test, then OPEN the pop-off.

## IV fluid rate

$\text{Dog kg} \times 5 \cdot \text{cat kg} \times 3\text{--}5$

mL/hr. Balanced crystalloid (Pardo 2024). Not 10 mL/kg/hr. Example:  $20 \times 5 = 100 \text{ mL/hr}$ .

| Patient                     | kg   | ASA | ETT                                | Bag / circuit               | IV fluids                | Fresh-gas flow                      |
|-----------------------------|------|-----|------------------------------------|-----------------------------|--------------------------|-------------------------------------|
| Example<br>20 kg dog        | 20   | 1   | 8.5 mm<br>8.0 / 9.0 ready          | 2 L circle<br>1.2 L → 2 L   | LRS 100 mL/hr<br>20 × 5  | 2–3 L/min, then<br>≥ 0.5 L/min      |
| Healthy Lab<br>elective OHE | 25   | 1   | 10 mm<br>9.5 / 10.5 ready          | 2 L circle<br>1.5 L → 2 L   | LRS 125 mL/hr<br>25 × 5  | 2–3 L/min, then<br>0.5–1 L/min      |
| MoMo<br>math only           | 4.25 | 4-E | 3.5–4.0 mm<br>cat; not dog formula | 0.5 L NRC<br>255 mL → 0.5 L | LRS 13 mL/hr<br>4.25 × 3 | 0.85–1.7 L/min<br>200–400 mL/kg/min |

# Surgical wound classification

## 1. Clean

No infection. Hollow viscus not entered.

Elective OHE, orchiectomy, simple skin mass.

Skip routine prophylaxis (Frey 2022).

## 2. Clean-contaminated

GI, respiratory, or urinary tract entered under control.  
Minor break in asepsis.

Cystotomy. Enterotomy without spillage.

## 3. Contaminated

Fresh open trauma. Major viscus spillage. Major break in sterile technique.

Time IV prophylaxis. Fix the break.

## 4. Dirty / infected

Pus, abscess, old trauma, perforated viscus with peritonitis.

Infected ear canal. Pyometra with pus.

This is treatment, not prophylaxis.

Write the class first. Clean elective: skip (Frey 2022 AAFP/AAHA stewardship). When prophylaxis is indicated: 30–60 min before incision; cefazolin 22 mg/kg IV (Gonzalez 2017, extra-label). Redose if surgery lasts >90 min (Whittem 1999). Stop at closure unless you are treating infection.

# Clip, dirty scrub, then the OR

## Patient and surgeon preparation

Clip and dirty-scrub in the prep area. Shoe covers, cap, and mask before the door. Full scrub, drape, and instruments in the OR.

|   |                                                                                                                  |   |                                                                                                 |   |                                                                                                    |   |                                                                                                           |    |                                                                                                                             |
|---|------------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Ready for prep</b><br>ETT in. Cuff holds. IV running. Surgical plane.<br>Then you may clip.                   | 2 | <b>Clip</b><br>#40 blade. Vacuum the hair. Express the bladder<br>if you will open the abdomen. | 3 | <b>Dirty scrub</b><br>Preliminary antiseptic in the prep area. This is not<br>the sterile prep.    | 4 | <b>Attire at the door</b><br>Shoe covers, hair cap, mask. Then enter the OR.<br>Gown is not hallway wear. | 5  | <b>Move and position</b><br>Into the OR. Pad. Ties. Pulse distal to each tie.<br>Final check.                               |
| 6 | <b>Full sterile scrub</b><br>In the room, after positioning. Clock contact time.<br>Spiral out. Drop the sponge. | 7 | <b>Gown, closed glove</b><br>Surgical hand scrub first. Hands stay in the cuffs.<br>Then glove. | 8 | <b>Drape</b><br>Four-quadrant towels, then the large drape. Only<br>the incision is in the window. | 9 | <b>Instrument setup</b><br>Open the pack. Check the indicator. Arrange.<br>Count sponges and instruments. | 10 | <b>Timeout. Mark incision</b><br>Announce the incision. Mark first cut on the<br>chart. Clock runs to the last skin suture. |

# Hair removal

Too narrow. Hair at the margin.



24 hours after OHE. Clip was wide enough.



Celiotomy: ~4 cm cranial to the xiphoid, ~4 cm caudal to the pubis, 3–4 cm past the nipples (Thieman Mankin 2019). A 4 kg cat does not have 20 cm of extra body wall. Wide enough to extend or place a drain. Warm the patient (Grubb: hypothermia <98 °F).

Photos: Uwe Gille, CC0; Liannadavis, CC BY-SA 4.0.

# Skin antiseptics

**Human Betadine Solution = 10% PVP-I. Veterinary paint = 5%, not 10%.  
Eye: 0.2% only. 2 min scrub + 2 min soak. Do not put 10% or 5% undiluted on the cornea (Roberts 1986).**

## CHG · Nolvasan

2% chlorhexidine acetate  
MIN contact: 2 to 4 min  
Keep out of eyes  
Trunk / intact skin  
Alcohol: rinse, then dry

## Skin · Betadine vet

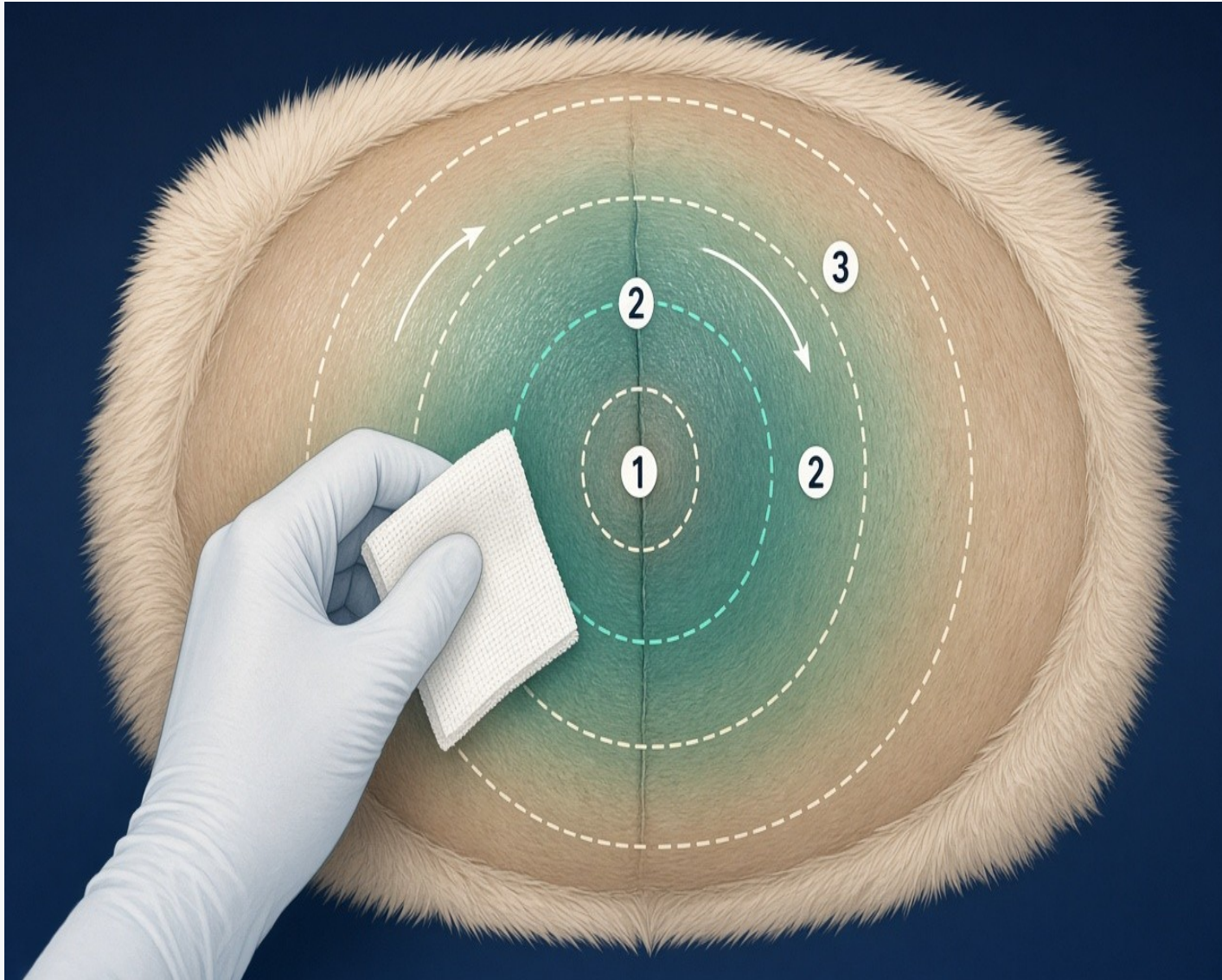
7.5% scrub  
MIN contact: about 5 min  
Rinse with sterile water  
Paint 5% solution  
Dry, then drape

## Eye · Roberts 1986

0.2%: 2 min scrub + 2 min soak  
10% stock → dilute 1:50  
5% bottle → dilute 1:25  
Not 10% or 5% undiluted  
Perforation: saline only

Nolvasan Surgical Scrub: 2% chlorhexidine acetate, DailyMed NDC 54771-8701, wash 2 to 4 min, keep out of eyes.  
BETADINE Surgical Scrub Veterinary NDC 67618-154 (7.5%, lather about 5 min). BETADINE Solution Veterinary NDC 67618-155 (5% paint, not 10%).  
Eye: Roberts 1986, healthy cornea. 1:50 of 10% = 0.2%, 2-min scrub + 2-min soak. Do not use undiluted 10% or undiluted 5% paint. 1:2 caused edema in 1/15 eyes. Perforation: saline.

# Patient skin preparation technique



## Technique

- ▶ Dirty (preliminary) scrub in the prep area, after the clip.
- ▶ Move to the OR. Position, pad, tie.
- ▶ Full (sterile) scrub in the room. Start at the incision. Spiral out. Drop the sponge.
- ▶ Clock the minimum contact time.
- ▶ Skin: 7.5% ~5 min, then 5% paint.
- ▶ Eye: 0.2% (10% 1:50 or 5% 1:25). 2 min + 2 min. Not 10% or 5% undiluted. Perforation: saline only.

# Patient positioning



## Visible in this photograph

- ET tube + pulse ox
- IV catheter + fluids
- Circle + reservoir bag
- V-trough / padding
- Ties snug; pulse distal to each tie
- Clip after a surgical plane

Tube tip: midway larynx to thoracic inlet (Grubb 2020).  
Confirm: ETCO<sub>2</sub>, bag move, no esophageal tube.

Photo: Anja, CC BY-SA 4.0.

# Draping

**OHE: hair visible at the drape edge**



**The window: clipped skin only**



Near towel first. Four towels box the field. Towel clamps on skin, not through the drape to viscera. Large drape last. Only the incision is in the window. Hair stays under the drape. Re-clip until the window is hair-free. Photos: Cpl. Samuel A. Nasso, USMC, public domain; MC1 Ryan D. McLearnon, USN, public domain.

# OR attire, gown, glove, instruments



Closed gloving: both hands remain inside the gown cuffs; the right cuff is used to pull the glove over the left hand.

## Before you cut

- ▶ Door: shoe covers, hair cap, mask. Then enter.
- ▶ Surgical hand scrub. Gown. Closed glove (hands stay in the cuffs).
- ▶ Drape. Then open the instrument pack. Check the indicator.
- ▶ Arrange. Count sponges and instruments. Mayo stand.
- ▶ Name the monitors: ECG, SpO2, ETCO2, temp, fluids.

Left: closed-gloving technique (hands stay inside the gown cuffs). Right: door attire, then gown, drape, open the pack, count, Mayo stand. Timeout. Announce the incision and mark it on the chart.

# Breaks in asepsis

## If you contaminate

Prep, gown, glove, drape: say it immediately. Re-glove, re-gown, or re-drape. Do not continue with a contaminated field.

After the incision: notice, announce, and fully correct. Sleeve in the abdomen, instrument off the table, hole in a glove: stop and fix it.

## Protect the patient

1. Control significant hemorrhage.
2. Create a secure abdominal wall closure.
3. Achieve and maintain an aseptic field.
4. Identify and protect adjacent organs before you clamp.
5. Complete a sponge and instrument count before closure.

Surgical competence includes the first 24 hours of physiologic monitoring and complication management (CPE MOA 2026 Surgery). A weak ligature or a weak linea is a technical failure you must recognize. That is not infinite legal liability for every recovery-ward or owner event.

# Knowledge check

**A**

**When do you clip the OHE field?**

After ready for prep. Clip and dirty-scrub in the prep area. Full sterile scrub in the OR after positioning.

**B**

**How do you prep conjunctiva with iodine?**

Intact cornea: 0.2% (10% 1:50 or 5% 1:25), 2 min scrub + 2 min soak. Not 10% or 5% undiluted.  
Perforation: saline only.

**C**

**Albumin 1.6 g/dL, elective mass. First move?**

Delay. Find the cause. Feed. Do not crystalloid-flood. Do not cut today.

**D**

**25 kg dog. Tube, bag, and IV fluid rate if you intubate?**

10 mm, 2 L circle, LRS 125 mL/hr (5 mL/kg/hr).

# Immediate recovery



## Recovery priorities

- ▶ Extubate when swallow returns.
- ▶ Most anesthetic deaths are in recovery, often in the first 3 hours (Grubb 2020).
- ▶ Pale + tachycardic after celiotomy: return to OR.
- ▶ NSAID: carprofen in the dog, Onsior in the cat. Do not stack with a steroid.
- ▶ E-collar on before they can lick.

# Anesthesia chart: start, incision, criteria

**Anesthesia chart**  
Mark anesthesia start and first incision. Record every 5–10 min against the range.

| What you plot                                                                                             | Normal operating range                      | Anesth. start                                                                                                                       | 5 min | Incision | 15 min                                                                                          | Last suture |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------|----------|-------------------------------------------------------------------------------------------------|-------------|
| Heart rate                                                                                                | dog 60–140 · cat 100–180                    |                                                                                                                                     |       |          |                                                                                                 |             |
| Respiratory rate                                                                                          | dog 8–20 · cat 10–25                        |                                                                                                                                     |       |          |                                                                                                 |             |
| BP SAP / MAP / DAP                                                                                        | SAP ≥90 · MAP ≥70 · DAP ≥40                 |                                                                                                                                     |       |          |                                                                                                 |             |
| Isoflurane or sevoflurane %                                                                               | to effect                                   |                                                                                                                                     |       |          |                                                                                                 |             |
| Oxygen flow L / min                                                                                       | 2–3 L, then 20–40 mL / kg / min · min 0.5 L |                                                                                                                                     |       |          |                                                                                                 |             |
| End-tidal CO2                                                                                             | 40–50 mm Hg · PPV if >60                    |                                                                                                                                     |       |          |                                                                                                 |             |
| SpO2                                                                                                      | ≥95%                                        |                                                                                                                                     |       |          |                                                                                                 |             |
| Temperature                                                                                               | ≥98 °F (36.7 °C)                            |                                                                                                                                     |       |          |                                                                                                 |             |
| Fluid rate mL / hr                                                                                        | dog 5 mL / kg / hr · cat 3–5                |                                                                                                                                     |       |          |                                                                                                 |             |
| Total fluid mL                                                                                            | rate × hours                                |                                                                                                                                     |       |          |                                                                                                 |             |
| Times you mark                                                                                            |                                             | Recording criteria                                                                                                                  |       |          | If out of range, then sign                                                                      |             |
| Anesthesia start. First incision. Last skin suture. Extubation. Write the clock time in the gold columns. |                                             | Every 5–10 min, in real time. Plot HR, RR, BP, inhalant, O2, ETCO2, SpO2, temp, fluids against the range. Write the number you see. |       |          | Event, time, and correction. Fluids: type, rate, total mL. Recovery analgesic. Sign the record. |             |

# Postop analgesia and incision care

## On-label NSAID

- ▶ Score pain. Dogs: Glasgow CMPS-SF. Cats: Feline Grimace Scale. Opioid as planned.
- ▶ Dog: carprofen (Rimadyl) 4.4 mg/kg SQ. Dogs only. Give ~2 hours before incision so levels are on board (pre-emptive; Rimadyl label).
- ▶ Cat: Onsior (robenacoxib) 2 mg/kg SQ. Cats only. First dose ~30 min before incision so levels are on board. OHE, castration, orthopedic. Up to 3 days.
- ▶ Dog: carprofen. Cat: Onsior. Do not swap species in this course.
- ▶ Do not stack an NSAID with a steroid. Skip NSAID if the patient is azotemic.

## Antibiotic injection and incision

- ▶ A postoperative antibiotic injection is treatment. Write the drug, dose, and why.
- ▶ Clean elective OHE: skip (Frey 2022). Do not send home 14 days of cephalexin.
- ▶ Dirty or infected: continue as therapy, not prophylaxis.
- ▶ Look twice daily: swelling, discharge, gapping, smell, heat.
- ▶ E-collar that stays on. Leash walks 14 days.
- ▶ Skin sutures typically 10–14 days.

# Complications in the first 24 hours

## Hemorrhage

Pale mm, tachycardia, distending abdomen, drip from incision, collapsing. Stabilize and return to OR.

## Airway / aspiration

Stertor, crackles, regurg on the pillow. Especially brachycephalics and after opioids.

## Hernia / evisceration

Linea failure. Emergency. Protect viscera with sterile moist towels, opioids, OR now.

## Dehiscence (later)

Often day 3–5. Skin versus fascia: fascial dehiscence is the emergency.

## SSI

Redness, pain, discharge, fever, usually after day 3. Culture. Open, lavage, and treat the organism you grow.

## Seroma / self-trauma

Dead space + motion. Restrict activity. Bandage if needed. Leave the fluid unless it is infected.

# Surgical report and discharge

## Minimum report elements

- ▶ Date/time, patient ID, surgeon, anesthesia, ASA.
- ▶ Procedure name and side. Position, clip/prep, approach.
- ▶ Findings in order. Ligatures, implants, samples.
- ▶ Suture: layer, material, size, pattern.
- ▶ Hemostasis, blood loss, sponge count, complications.
- ▶ Postop plan: fluids, pain, feeding, recheck.

## Discharge, verbal and written

- ▶ What we did, in one sentence.
- ▶ When to start food and water, and how much.
- ▶ Every medication: name, dose, time, with food?
- ▶ Incision care and 14-day leash walks.
- ▶ E-collar on except when directly watching.
- ▶ Next appointment. ER criteria. After-hours number.

# Emergency criteria for discharge

## Come now

Collapse. Pale gums.  
Distended abdomen.  
Unstoppable bleeding. Can't breathe. Can't urinate.  
Evisceration. Uncontrolled pain.

## Call today

Not eating by morning. A few vomits. Mild incision redness. Diarrhea. E-collar problems.

## Expected

Sleepy tonight. Small bruise. Slightly tacky incision. Reduced appetite this evening.

# Answer: MoMo is Status 4-E

MoMo · SF DSH · 4.25 kg · today's PE and today's labs write the status. Do not cut.

| Disease                                                                                                                                 | Uncompensated                                                                                        | Not Status 3                                                                                                      | Do not cut                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Acute vomiting and lethargy. Azotemia. Right kidney fluid-filled and non-functional. Left kidney reduced corticomedullary architecture. | BUN 48.7 → 100. Cr 3.0 → 4.71. BE −7.4. Na 142. HR 200, tacky mm. That is a constant threat to life. | Status 3 is moderate and compensated. A HCT of 44% in range does not write Status 1. This is not compensated CKD. | Exploratory cancelled. NSAIDs contraindicated. Owner elected euthanasia. Record SURGERY CANCELLED. |

# Answer: Willie is Status 3-E

Willie · Cavalier · 13.7 kg · today's PE and today's labs write the status

| Disease                                                                                                                                           | Compensated                                                                               | Not Status 1                                                                               | Not Status 4                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| AS otitis. Head tilt left. Circling left. Nystagmus rapid left, slow right. Wheelbarrow right absent. Knuckling, right limbs. Grade II/VI murmur. | Pink, CRT 2 s. BUN 11.9 and creatinine 0.7 in range. Not a constant threat to life today. | HCT 47.5% in range does not write Status 1. This neuro PE is not a normal healthy patient. | Phos 6.0 and glucose 130 are high. They do not write Status 4. Status 4 is a constant threat to life. |

# Key points

1

Pre-anesthetic assessment: TPR, mm/CRT, then PCV, TP, BUN. Interpret. Circle ASA. Then IM or SQ premed.

2

Write drug, concentration, dose, volume, and route. Then IV fluid name and rate, ETT size, circuit, and fresh-gas flow.

3

Ready for surgical prep: airway secured, machine on, patent IV running, surgical plane, monitoring started.

4

Patient: clip, dirty scrub, move, full scrub, drape. Surgeon: shoe covers, cap, mask, then gown and closed glove. Instrument table, then announce incision.

5

Mark anesthesia start and first incision. Fill the chart every 5–10 min against the operating range. Then complications, recovery analgesics, and sign.

6

First 24 hours: monitor and manage complications. A weak ligature or a weak linea is still your technical problem.

# Questions

When is the patient ready for clippers?

What do you write before the first IM or SQ premed?

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